i.e $10 = \frac{230}{R}$

$R = \frac{230}{10} = 23\Omega$

4) *An electrical circuit is having 40Ω resistance value. When 6A current is following through this circuit, find the voltage?*

Resistance (R)= 40Ω
Current (I) = 6A
Voltage (V) = ?
According to ohm's law, $I = \frac{V}{R}$
i.e $6 = \frac{V}{40}$
Voltage (V) = 6 × 40 = 240V.

2.6 TYPES OF CIRCUITS

The circuit is defined as the current flowing from the supply points through the load to complete the path. In this chapter, we will study about the types of the electrical circuits. There are three types of electrical circuit namely;

i) Closed circuit,
ii) Open circuit and
iii) Short circuit.

These are explained below

i) Closed Circuit

When loads are connected in series between two terminals of electric supply, in such a way that the current passing through the load is to activate the circuit. It is called as closed circuit.

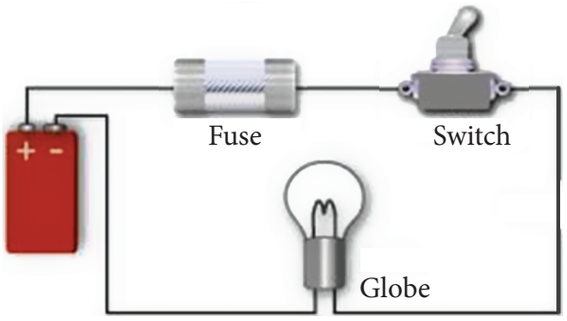


Fig 2.2 *Closed Circuit*

Example:- In our home, each room has different wiring circuit i.e lighting circuit, power circuit, etc.

For example, if we switch ON the light circuit, the light will glow, it is called a closed circuit.

ii) Open Circuit

In this circuit, if there is no way to the flow of current due to disconnection of wire or the switch is in OFF condition, then the circuit is called open circuit.

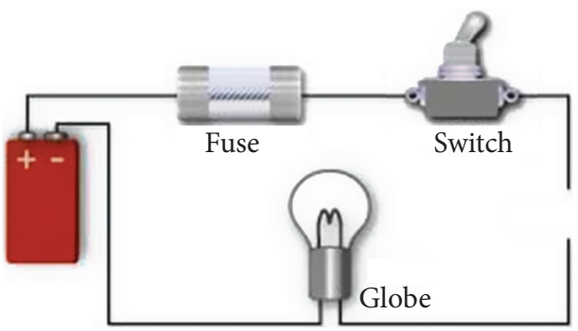


Fig 2.3 *Open Circuit*

Ex: In lighting circuit, if the switch is in OFF condition or the wire is broken in any place the lamp will not glow and it is called an open circuit.

iii) Short Circuit

In this circuit, the two terminals of the supply is connected directly without a load and the current flow is infinite

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